Day 7 - Sum of First N Natural Numbers

Question

Ek number **N** diya gaya hai.

1 se N tak sabhi numbers ka sum find karo.

Example

Input

```
N = 5
```

Output

```
15
```

Explanation

```
1 + 2 + 3 + 4 + 5 = 15
```

Approach 1 - Using Loop

Logic

1 se N tak loop chalao aur sum me add karte jao.

Dry Run

```
N = 5

sum = 0

sum = 0 + 1 = 1
sum = 1 + 2 = 3
sum = 3 + 3 = 6
sum = 6 + 4 = 10
sum = 10 + 5 = 15
```

Java Code

```
public class Day7 {
    public static void main(String[] args) {

        int n = 5;
        int sum = 0;

        for(int i = 1; i <= n; i++) {
            sum += i;
        }

        System.out.println(sum);
    }
}
```

Time Complexity

```
O(N)
```

Space Complexity

```
O(1)
```

Approach 2 - Using While Loop

Java Code

```
public class Day7 {
    public static void main(String[] args) {

        int n = 5;
        int i = 1;
        int sum = 0;

        while(i <= n) {
            sum += i;
            i++;
        }

        System.out.println(sum);
    }
}
```

Time Complexity

O(N)

Space Complexity

O(1)

Approach 3 - Using Recursion

Logic

```
sum(n) = n + sum(n-1)
```

Example

```
sum(5)

5 + sum(4)

5 + 4 + sum(3)

5 + 4 + 3 + sum(2)

5 + 4 + 3 + 2 + sum(1)

5 + 4 + 3 + 2 + 1

= 15
```

Java Code

```
public class Day7 {

    static int sum(int n) {

        if(n == 1)
            return 1;

        return n + sum(n - 1);
    }

    public static void main(String[] args) {

        int n = 5;

        System.out.println(sum(n));
    }
}
```

Time Complexity

```
O(N)
```

Space Complexity

```
O(N)
```

(Call Stack ki wajah se)

Approach 4 - Formula Method ✅ (Optimal)

Logic

Mathematical Formula:

```
Sum = n × (n + 1) / 2
```

Example

```
N = 5

= 5 × 6 / 2

= 30 / 2

= 15
```

Java Code

```
public class Day7 {
    public static void main(String[] args) {

        int n = 5;

        int sum = n * (n + 1) / 2;

        System.out.println(sum);
    }
}
```

Time Complexity

```
O(1)
```

Space Complexity

```
O(1)
```

Interview Follow-Up Questions

Q1. Best Solution?

Formula Method

```
n * (n + 1) / 2
```

Kyuki loop hi nahi chal raha.

Q2. Agar N = 100000 ho to?

Loop:

```
100000 iterations
```

Formula:

```
Sirf 1 calculation
```

Isliye Formula better hai.

Q3. Formula kisne diya tha?

Ye famous story hai:

Carl Friedrich Gauss ne bachpan me 1 se 100 ka sum instantly nikala tha.

```
1 + 100 = 101
2 + 99 = 101
3 + 98 = 101
```

50 pairs × 101 = 5050

Interview Concept of the Day

Recurrence Relation

Recursion me:

```
f(n) = n + f(n-1)
```

Isko Recurrence Relation bolte hain.

Future me:

- Recursion
- DP
- Trees

sab me use hoga.

Summary

Approach	Time	Space
For Loop	O(N)	O(1)
While Loop	O(N)	O(1)
Recursion	O(N)	O(N)
Formula	O(1)	O(1)

Practice Questions

Find Sum:

1

```
N = 10
```

2

```
N = 50
```

3

```
N = 100
```

4

```
N = 1000
```

5

```
N = 5000
```

Formula se solve karo.